

Laptop Price Prediction using Machine Learning

Internship Project Report

Student Name:

Sonia

Internship Domain:

Artificial Intelligence & Machine Learning (AI/ML)

Task Number: Task 1

Project: Laptop Price Prediction using Machine Learning

Tools & Technologies:

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Introduction

This project focuses on Python programming, data analysis, exploratory data analysis, statistical analysis, and data visualization using a real-world laptop dataset. The dataset was analyzed using Python in Google Colab to understand its structure, clean the data, examine important features, calculate statistical measures, create visualizations, and generate meaningful insights. The project follows a complete data-analysis workflow from dataset loading and understanding to visualization and final conclusions.

1.Dataset Name

Laptop _Data

2. Dataset Source

Source: Kaggle Laptop Dataset.

3. Dataset Description

The laptop dataset contains information about different laptops from various companies. It includes both categorical and numerical features that can be analyzed to understand laptop characteristics and pricing patterns. The dataset is suitable for exploratory data analysis, statistical analysis, and visualization.

4. Dataset Size

The original laptop dataset contains 1,303 records and 12 columns. During data cleaning, the unnecessary Unnamed: 0 column was removed because it does not represent a meaningful laptop feature. After cleaning, the dataset contains 1,303 records and 11 relevant features for analysis.

5. Features / Columns Description

The dataset contains several features describing laptop specifications and characteristics, including company, laptop type, RAM, weight, and price. The features include both categorical and numerical variables, making the dataset suitable for comparison, statistical analysis, and visualization.

6. Methodology

The project followed a structured data-analysis workflow. First, the required Python libraries were imported and the laptop dataset was loaded into a Pandas DataFrame. The dataset was then explored to understand its structure, columns, data types, and values. Data cleaning was performed to handle missing values and remove unnecessary columns. Exploratory data analysis and statistical analysis were then performed, followed by multiple visualizations to identify patterns and relationships. Finally, the results were interpreted to generate meaningful findings and insights.

7. Data Loading

The laptop dataset was loaded into a Pandas DataFrame using the `pd.read_csv()` function in Google Colab. The `df.head()` function was then used to verify that the dataset had been loaded successfully and to view the first few records.

The screenshot shows a Google Colab notebook titled 'Task_01_Data_Analysis.ipynb'. The code cell contains the following Python code:

```
df = pd.read_csv("laptop_data.csv")  
  
df.head()
```

The output of `df.head()` is a DataFrame with 12 columns: Unnamed: 0, Company, TypeName, Inches, ScreenResolution, Cpu, Ram, Memory, Gpu, OpSys, Weight, and Price. The DataFrame contains 5 rows of data.

	Unnamed: 0	Company	TypeName	Inches	ScreenResolution	Cpu	Ram	Memory	Gpu	OpSys	Weight	Price
0	0	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x1600	Intel Core i5 2.3GHz	8GB	128GB SSD	Intel Iris Plus Graphics 640	macOS	1.37kg	71378.6832
1	1	Apple	Ultrabook	13.3	1440x900	Intel Core i5 1.8GHz	8GB	128GB Flash Storage	Intel HD Graphics 6000	macOS	1.34kg	47895.5232
2	2	HP	Notebook	15.6	Full HD 1920x1080	Intel Core i5 7200U 2.5GHz	8GB	256GB SSD	Intel HD Graphics 620	No OS	1.86kg	30636.0000
3	3	Apple	Ultrabook	15.4	IPS Panel Retina Display 2880x1800	Intel Core i7 2.7GHz	16GB	512GB SSD	AMD Radeon Pro 455	macOS	1.83kg	135195.3360
4	4	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x1600	Intel Core i5 3.1GHz	8GB	256GB SSD	Intel Iris Plus Graphics 650	macOS	1.37kg	96095.8080

The notebook interface also shows the 'df.info()' command in the next cell. The Windows taskbar at the bottom indicates the time is 7:01 PM on 8/7/2026.

8. Data Cleaning

Data cleaning was performed before detailed analysis. Missing values were checked using `df.isnull().sum()`, and an unnecessary Unnamed: 0 column was identified and removed because it did not provide a meaningful feature for the analysis

Task_01_Data_Analysis.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

df.isnull().sum()

```

0
Unnamed: 0    0
Company       0
TypeName      0
Inches        0
ScreenResolution  0
Cpu           0
Ram           0
Memory        0
Gpu           0
OpSys         0
Weight        0
Price         0
dtype: int64

```

Activate Windows
Go to Settings to activate Windows.

Variables Terminal

Type here to search

28°C Mostly sunny 7:02 PM 8/7/2026

9. Exploratory Data Analysis

Exploratory data analysis was performed using functions such as `df.head()`, `df.info()`, and `df.describe()`. These functions helped examine the dataset structure, column names, data types, non-null values, and summary statistics before performing further analysis.

```

1 df.info()
...
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1303 entries, 0 to 1302
Data columns (total 12 columns):
 #   Column                Non-Null Count  Dtype
---  ---
 0   Unnamed: 0            1303 non-null  int64
 1   Company               1303 non-null  object
 2   TypeName              1303 non-null  object
 3   Inches                1303 non-null  float64
 4   ScreenResolution      1303 non-null  object
 5   Cpu                   1303 non-null  object
 6   Ram                   1303 non-null  object
 7   Memory                1303 non-null  object
 8   Gpu                   1303 non-null  object
 9   OpSys                 1303 non-null  object
10  Weight                1303 non-null  object
11  Price                 1303 non-null  float64
dtypes: float64(2), int64(1), object(9)
memory usage: 122.3+ KB

```

df.describe()

...	Unnamed: 0	Inches	Price
count	1303.00000	1303.000000	1303.000000
mean	651.00000	15.017191	59870.042910
std	376.28801	1.426304	37243.201786
min	0.00000	10.100000	9270.720000
25%	325.50000	14.000000	31914.720000
50%	651.00000	15.600000	52054.560000
75%	976.50000	15.600000	79274.246400
max	1302.00000	18.400000	324954.720000

13. GitHub Repository

<https://github.com/sonia-kanwal>

OBJECTIVE

The main objective of this project is to build a Machine Learning model that can predict the price of a laptop based on its specifications. The project also aims to understand the complete machine learning workflow, including data loading, data cleaning, data visualization, model training, and model evaluation.

Problem Statement

Laptop prices depend on many factors such as the company, processor, RAM, storage, screen size, graphics card, and operating system. It is difficult to estimate the price accurately without analyzing these features. This project uses historical laptop data to train a Machine Learning model that predicts laptop prices automatically.

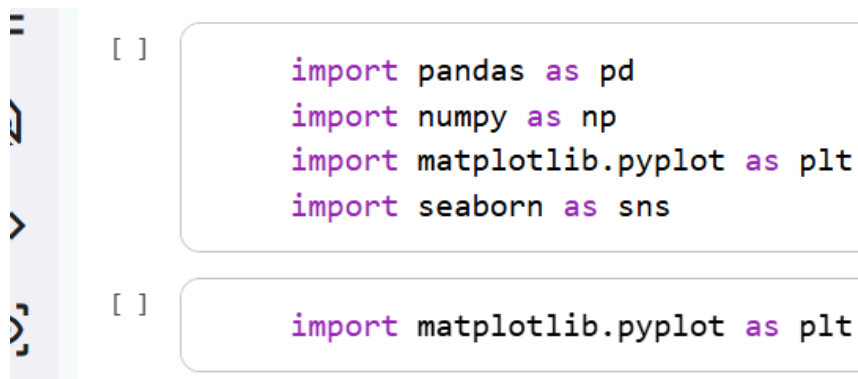
Technologies Used

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Linear Regression

Implementation Steps

Step 1: Import Libraries

Imported the required Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn for data analysis, visualization, and machine learning.



```
[ ] import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

[ ] import matplotlib.pyplot as plt
```

Figure:python libraries import for data analysis and visualization

Step 2: Load Dataset

The laptop dataset was loaded into a Pandas DataFrame using `pd.read_csv()`. The `df.head()` function was then used to view the first few records of the dataset.

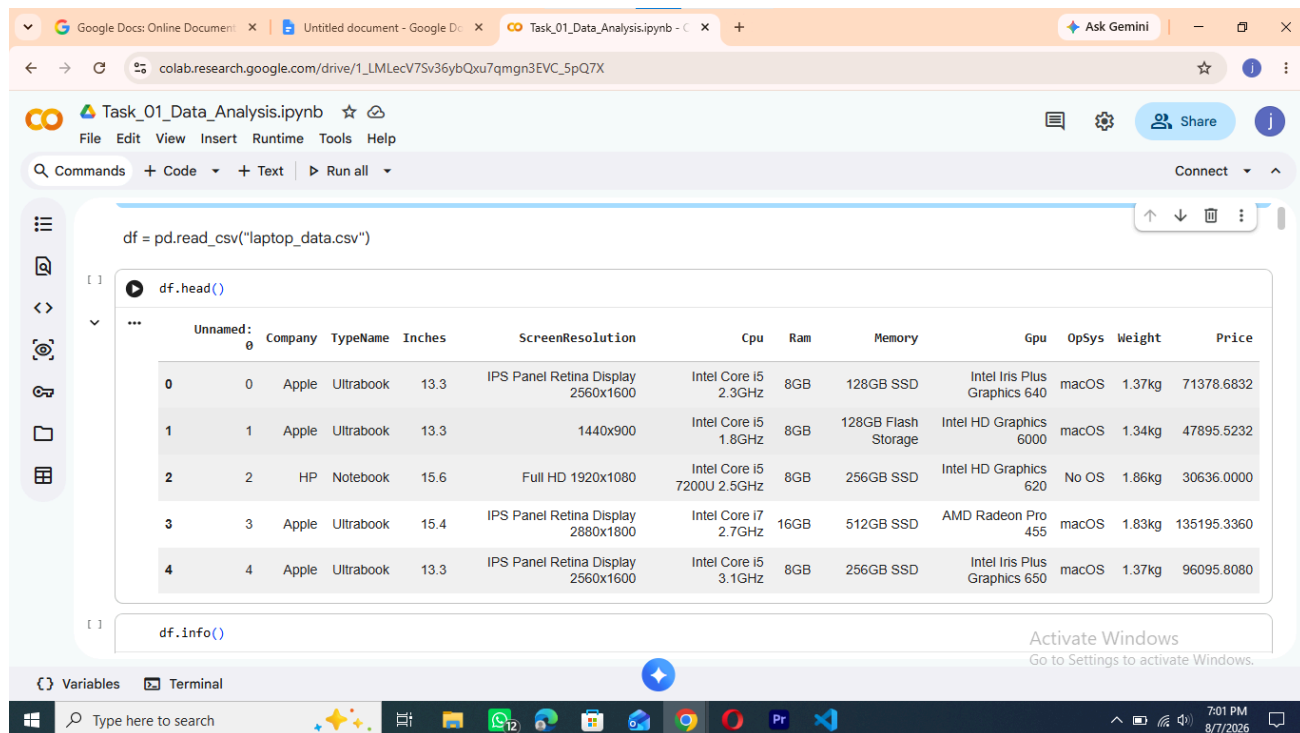


Figure:loading and previewing the laptop dataset

Step 3: Explore the Dataset

The dataset was explored using `df.head()`, `df.info()`, and `df.describe()`. These functions helped understand the dataset structure, column names, data types, non-null values, and summary statistics.


```

1 df.info()
... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 1303 entries, 0 to 1302
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Unnamed: 0            1303 non-null   int64
1   Company               1303 non-null   object
2   TypeName              1303 non-null   object
3   Inches                1303 non-null   float64
4   ScreenResolution      1303 non-null   object
5   Cpu                   1303 non-null   object
6   Ram                   1303 non-null   object
7   Memory                1303 non-null   object
8   Gpu                   1303 non-null   object
9   OpSys                 1303 non-null   object
10  Weight                1303 non-null   object
11  Price                 1303 non-null   float64
dtypes: float64(2), int64(1), object(9)
memory usage: 122.3+ KB

```

Figure 1:Dataset information and data types

```

df.describe()
...

```

	Unnamed: 0	Inches	Price
count	1303.000000	1303.000000	1303.000000
mean	651.000000	15.017191	59870.042910
std	376.28801	1.426304	37243.201786
min	0.000000	10.100000	9270.720000
25%	325.500000	14.000000	31914.720000
50%	651.000000	15.600000	52054.560000
75%	976.500000	15.600000	79274.246400
max	1302.000000	18.400000	324954.720000

Figure 2:Descriptive statics of the dataset

Step 4: Data Cleaning

The dataset was checked for missing values using `df.isnull().sum()`. The unnecessary `Unnamed: 0` column was then removed to keep only the relevant features for analysis and model training.

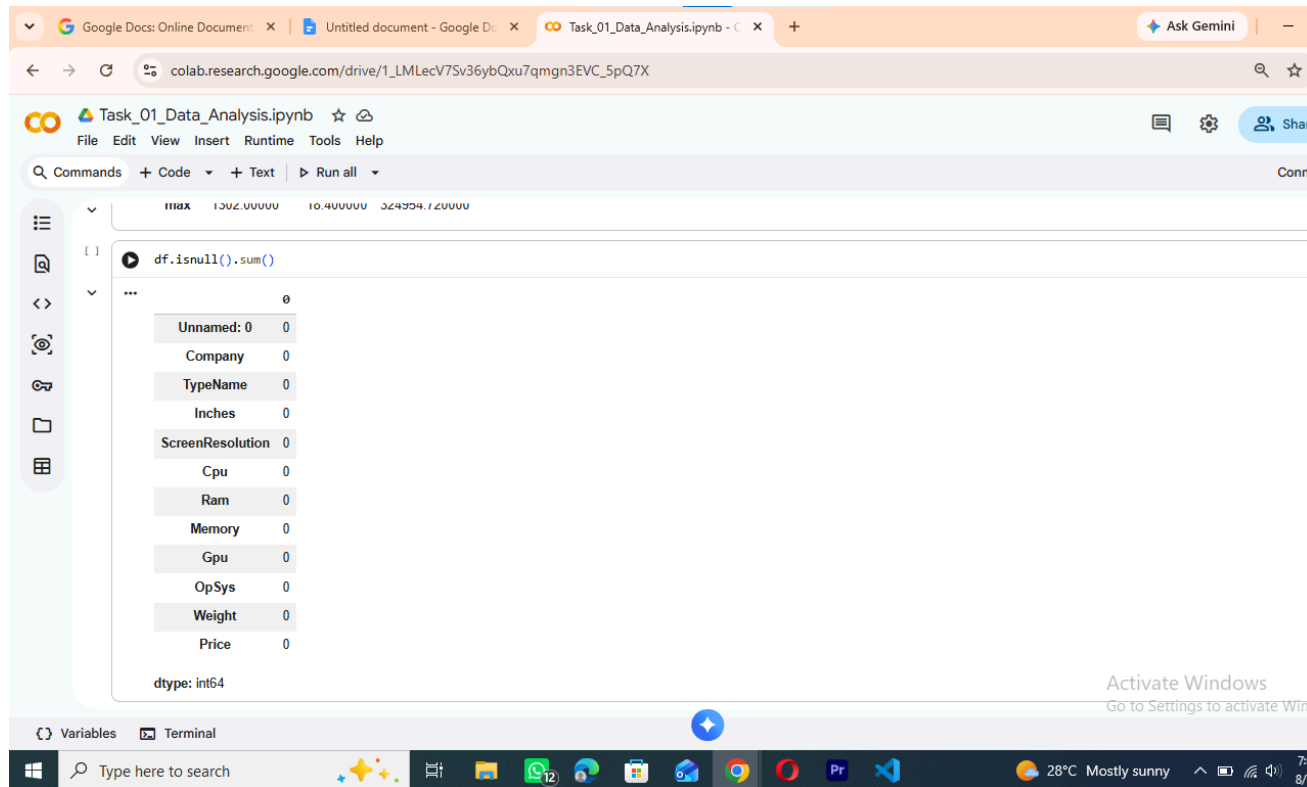


Figure 01: Missing values check

```
print(df.columns.tolist())

df = df.drop(columns=["Unnamed: 0"], errors="ignore")

print(df.columns.tolist())

['Unnamed: 0', 'Company', 'TypeName', 'Inches', 'ScreenResolution', 'Cpu', 'Ram', 'Memory', 'Gpu', 'OpSys', 'Weight', 'Price']
['Company', 'TypeName', 'Inches', 'ScreenResolution', 'Cpu', 'Ram', 'Memory', 'Gpu', 'OpSys', 'Weight', 'Price']
```

Figure 2: Removal of unnecessary column

Data Visualization:

Data visualization was performed to understand patterns and relationships within the laptop dataset. Different graphs were created using Matplotlib and Seaborn to analyze laptop brands, laptop types, and other important features.

Graph 1 — Laptop Distribution by Company

This graph shows the number of laptops available from different companies in the dataset. It helps compare the representation of each laptop company.

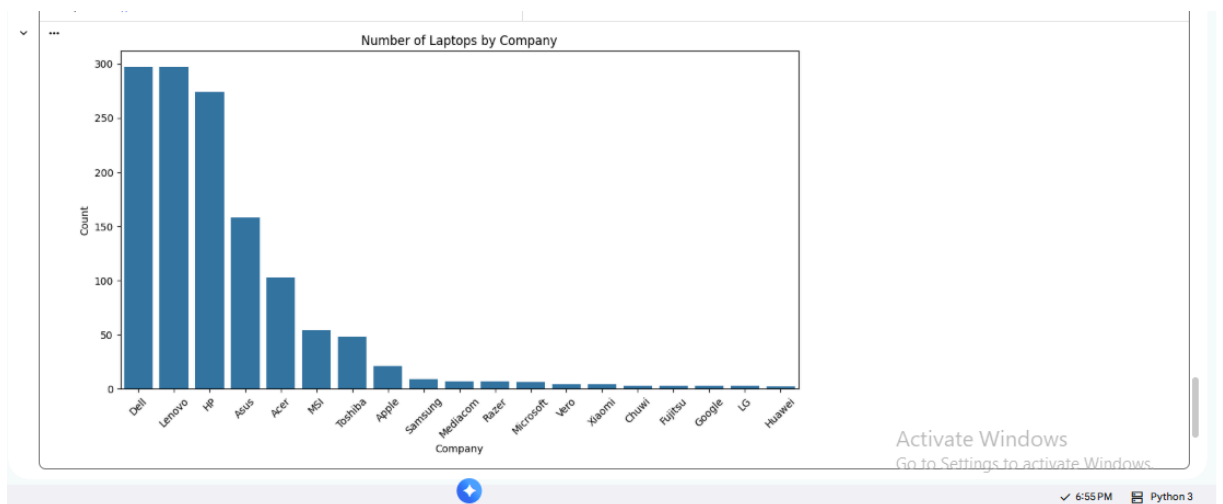


Figure: Distribution of laptop by computer

Graph 2 — Laptop Type Distribution

This graph shows the distribution of different laptop types in the dataset. It helps identify which types of laptops are most common and which have fewer records.

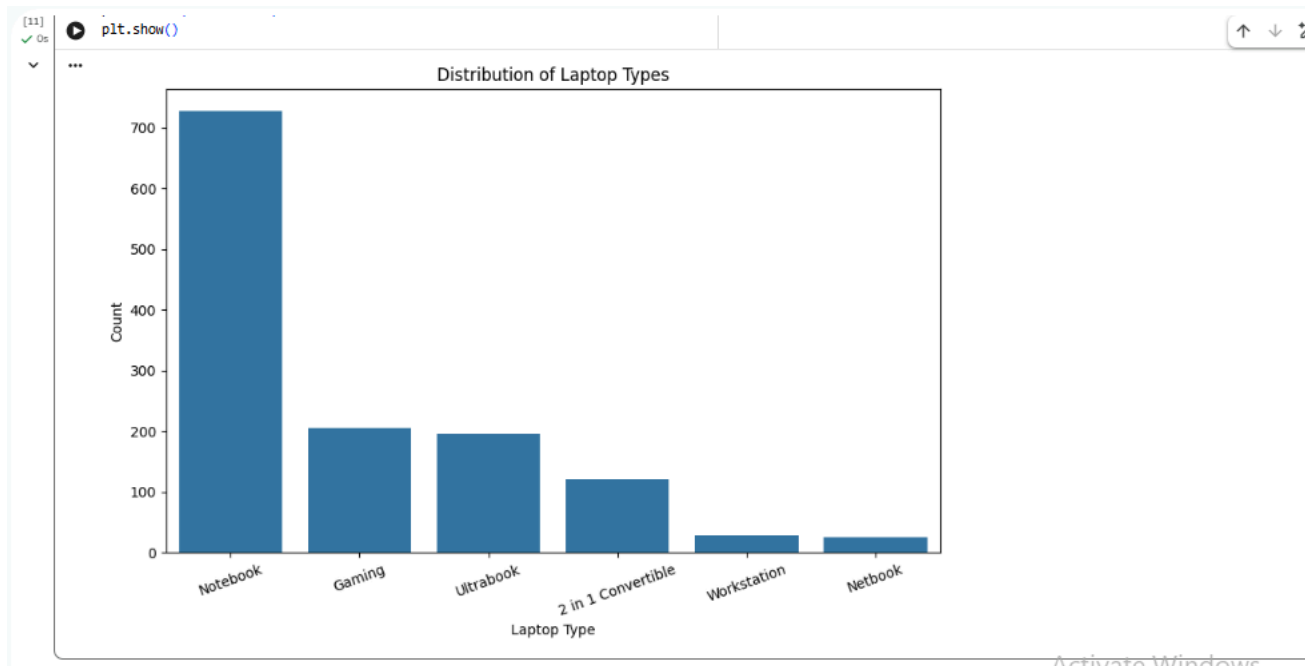


Figure: Distribution of laptops by type

Graph 3 — Laptop Price Distribution

This graph shows the distribution of laptop prices in the dataset. It helps visualize how laptop prices are spread across different price ranges and provides an overview of the pricing pattern.

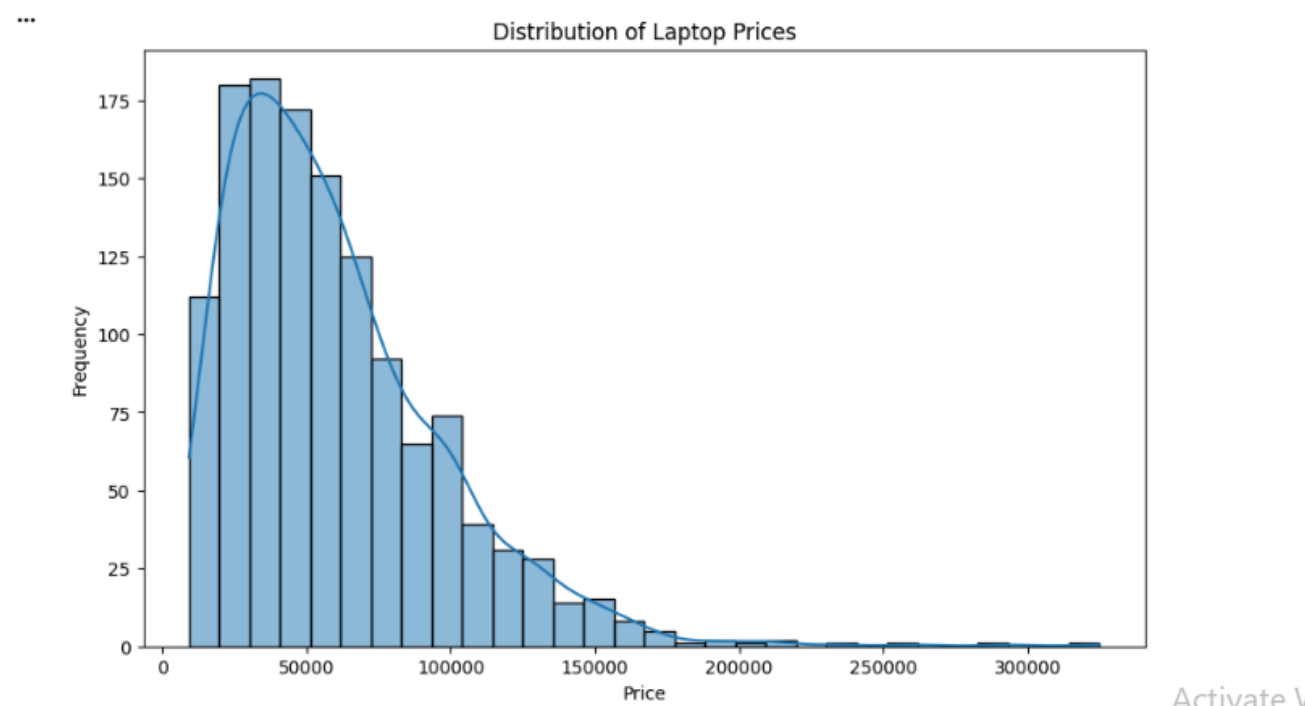


Figure:Distribution of laptop prices

Graph 4 — Correlation heatmap

Correlation Heatmap: The correlation heatmap was used to examine relationships between numerical variables in the dataset. It helps identify which features have stronger positive or negative relationships with laptop price and with each other.

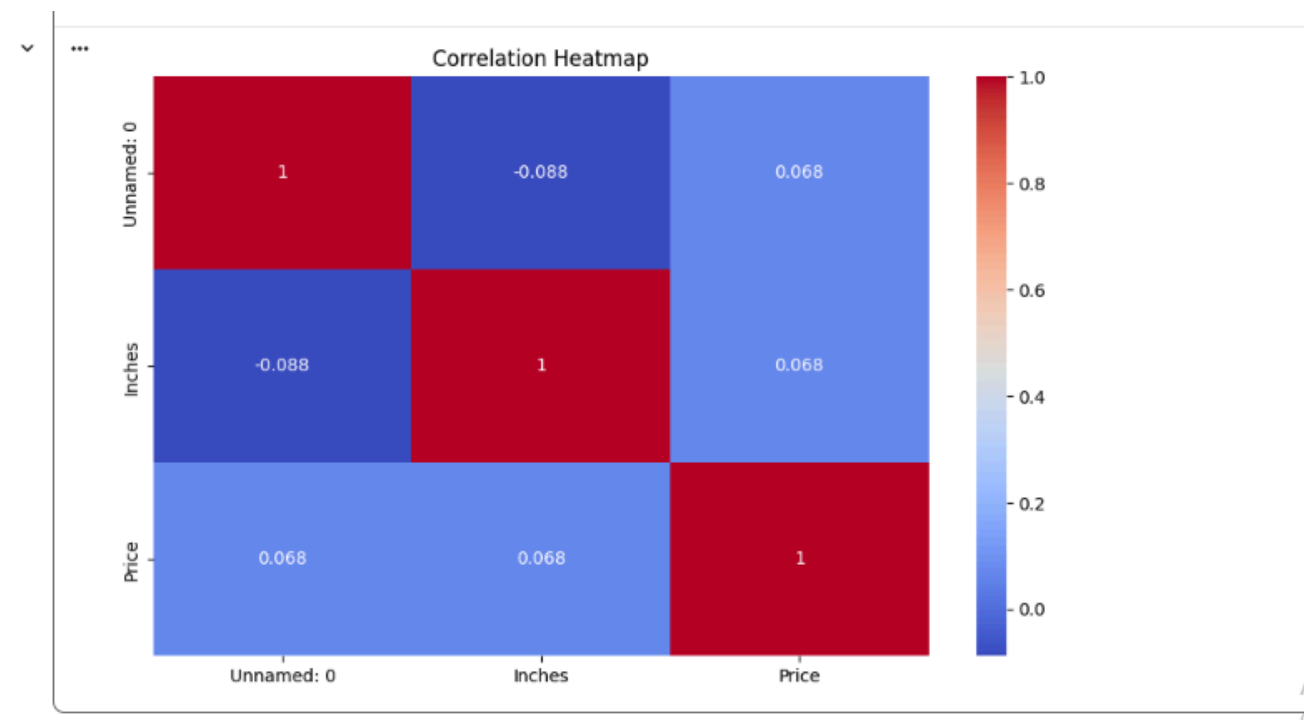
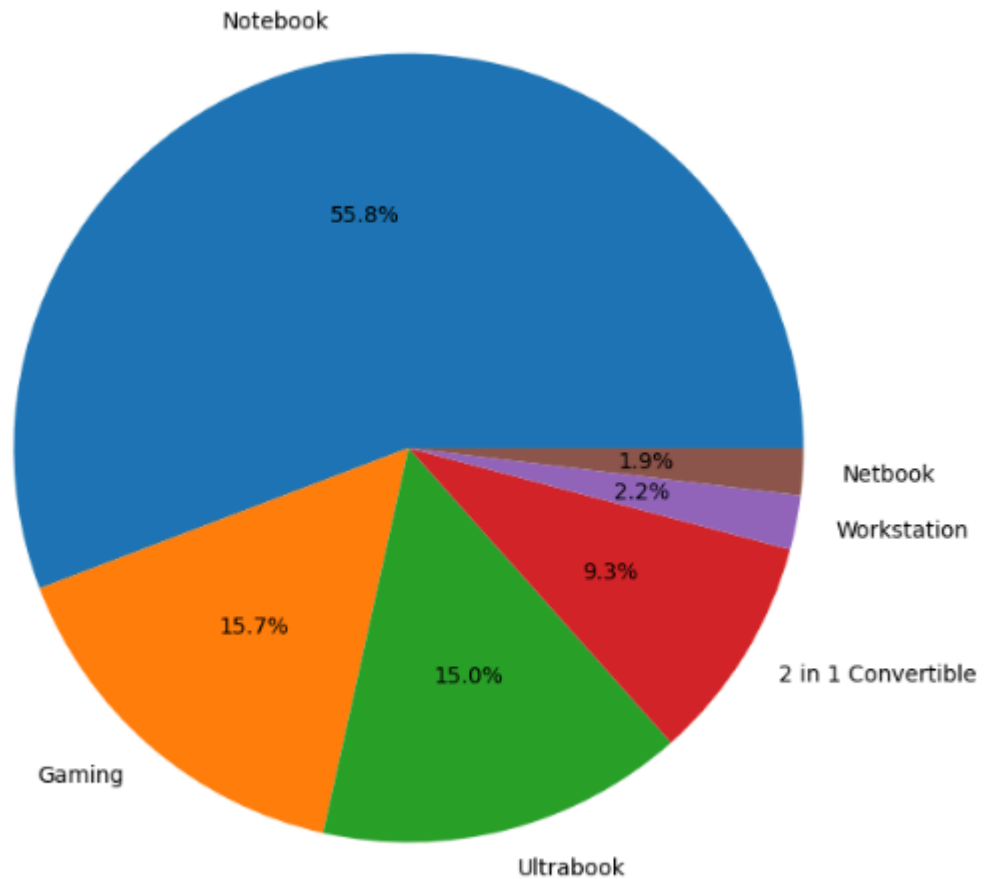


Figure: correlation heatmap showing relationship between numerical features in the laptop

Graph 5 — Pie Chart

The pie chart was used to show the proportion of different laptop types in the dataset. It provides a clear visual comparison of how much each laptop type contributes to the overall dataset.



RAM vs Laptop Price

Figure: pie chart showing the proportion of different laptop types in the dataset

Graph 6 —Scatter Plot

The scatter plot was used to examine the relationship between RAM and laptop price. Each point represents a laptop, allowing the relationship between RAM capacity and price to be visually examined.

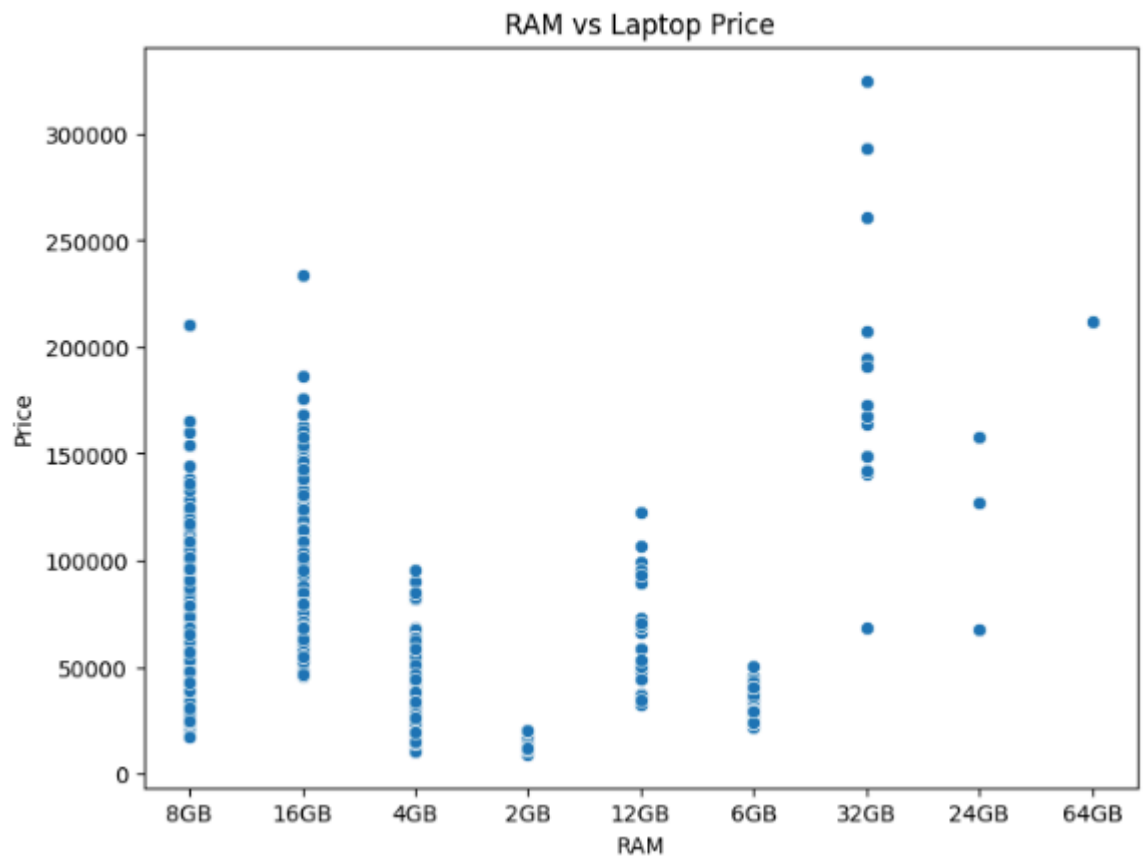


Figure: scatter plot showing the relation between RAM and laptop price

Graph 7 — Line Chart

The line chart was used to compare the average laptop prices across different companies. It helps visualize changes in average price between companies and makes differences in pricing easier to observe.

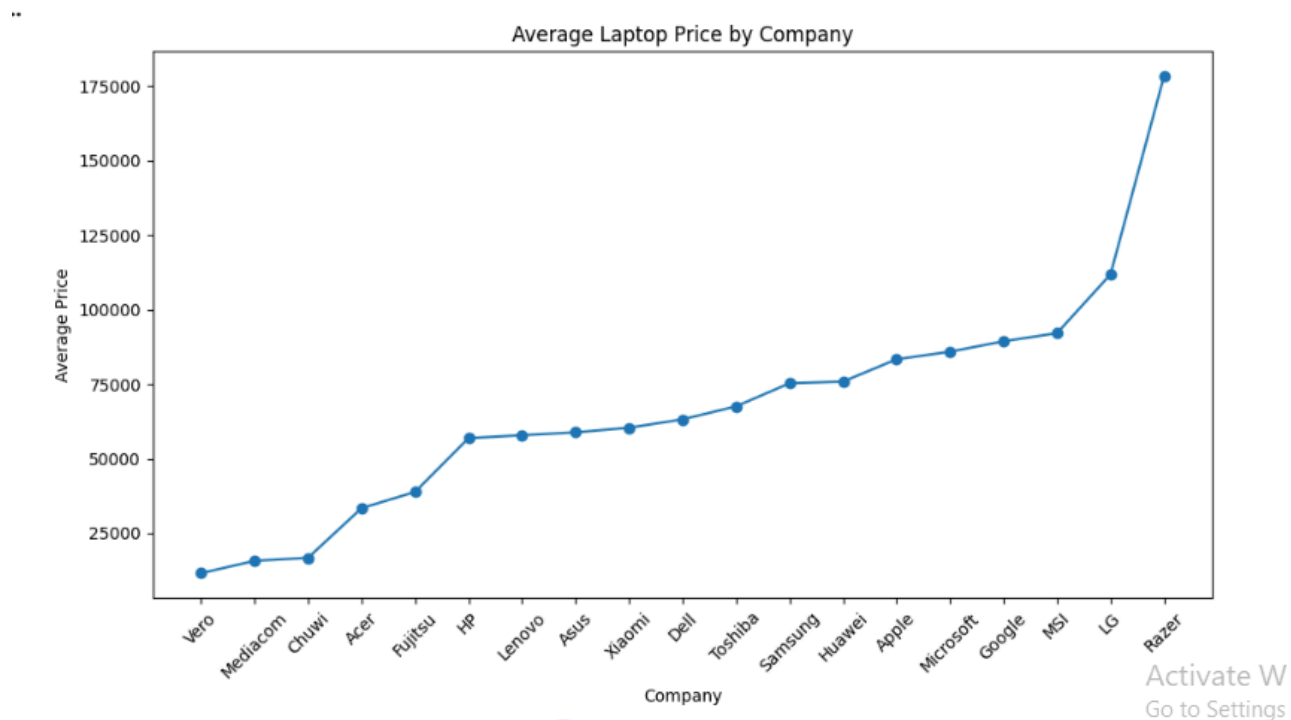


Figure: Line Chart showing the average laptop price by computer

Graph 8 — Box Plot

The box plot was used to examine the spread of laptop prices and identify the central tendency and possible outliers. It provides a compact view of how the prices are distributed and how much they vary.

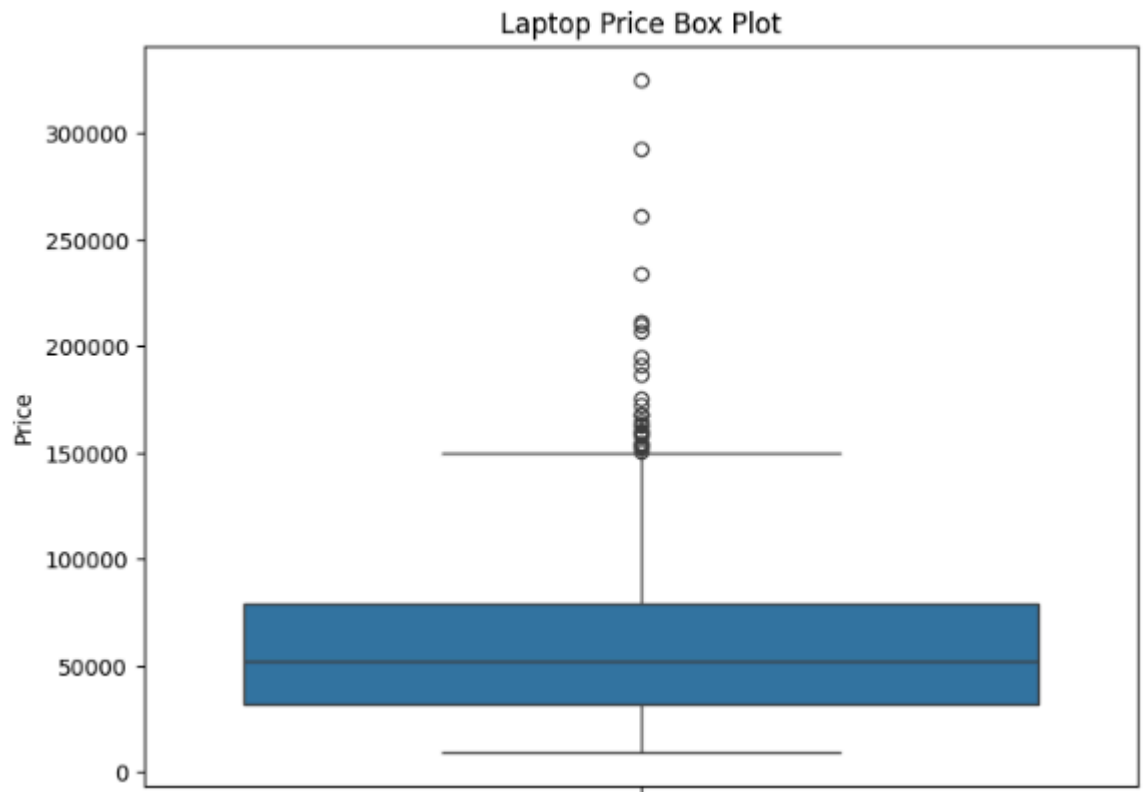


Figure: Box plot showing the distribution and spread of laptop prices

Graph 9 — Count Plot

The count plot was used to compare the number of laptops belonging to each laptop type. It helps identify which laptop categories are most common in the dataset.

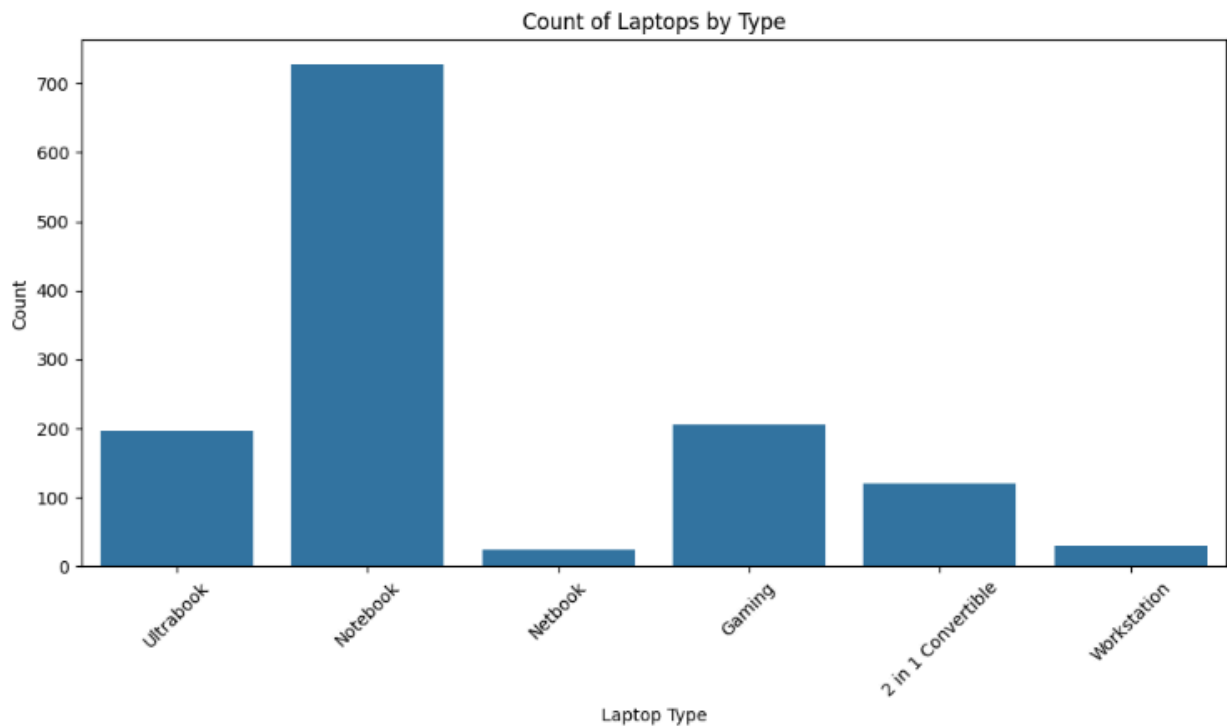


Figure: count plot showing the number of laptops in each laptop type category

Graph 10 — Pair Plot

The pair plot was used to examine relationships and patterns between multiple numerical variables at the same time. It combines scatter plots and distributions to provide an overall view of the relationships in the dataset.

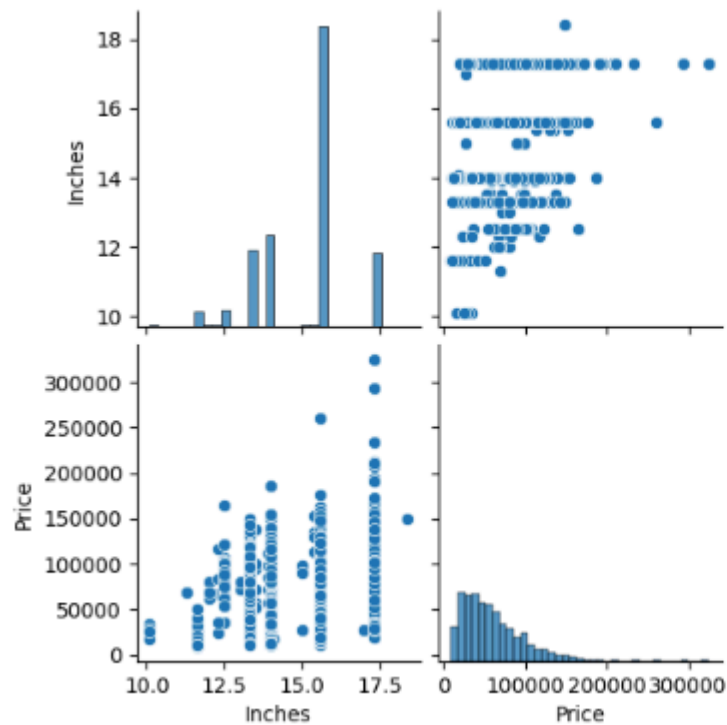


Figure: pair plot showing relationships between selected numerical variables

Graph 11 — Violin Plot

The violin plot was used to compare the distribution and spread of laptop prices across different laptop types. It helps show where prices are concentrated and how the price distributions differ between categories.

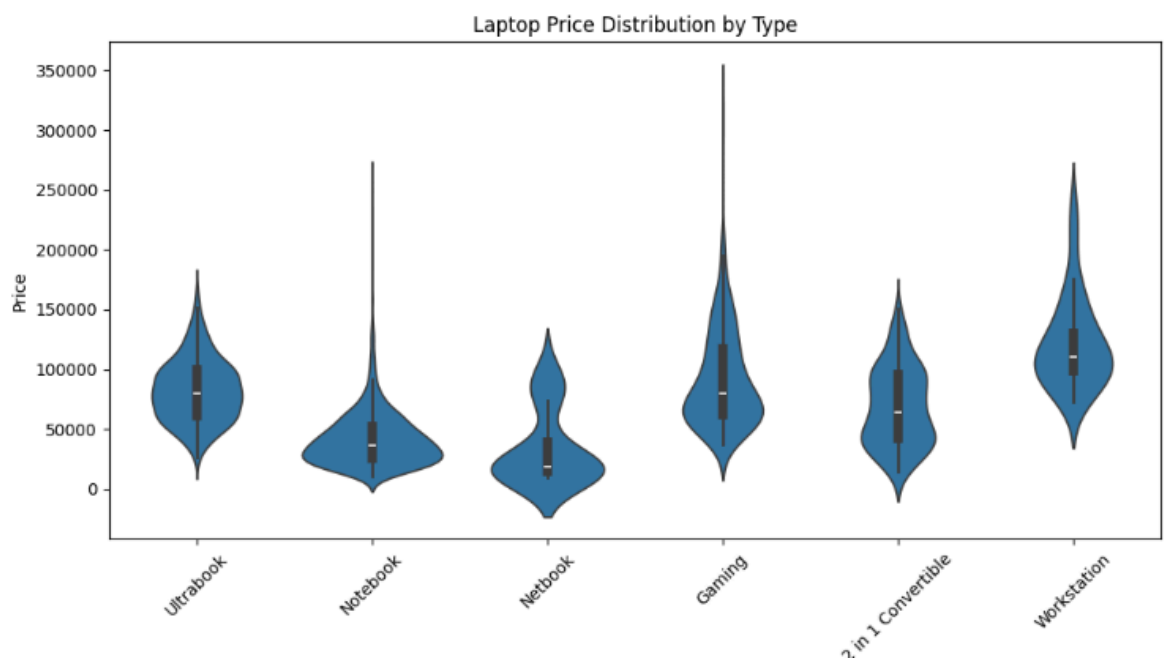


Figure: violin plot showing laptop price distribution across laptop types

Statistical Analysis

Statistical analysis was performed to understand the distribution and variation of laptop prices. The average laptop price was approximately 59,870, while the median price was approximately 52,055. The standard deviation was approximately 37,243, indicating considerable variation in laptop prices. The minimum recorded price was approximately 9,271, while the maximum price was approximately 324,955. The higher mean compared with the median, together with the high maximum price, indicates that some higher-priced laptops influence the overall price distribution.

```
# Statistical Analysis

print("Mean Price:", df['Price'].mean())
print("Median Price:", df['Price'].median())
print("Standard Deviation:", df['Price'].std())
print("Minimum Price:", df['Price'].min())
print("Maximum Price:", df['Price'].max())

*** Mean Price: 59870.042910207216
    Median Price: 52054.56
    Standard Deviation: 37243.20178619774
    Minimum Price: 9270.72
    Maximum Price: 324954.72
```

Figure: statistical analysis of laptop prices

Major Findings

The exploratory analysis provided several important findings about the laptop dataset. Laptop prices showed considerable variation, with prices ranging from approximately 9,271 to 324,955 and an average price of approximately 59,870. The median price was approximately 52,055, indicating that the distribution is influenced by higher-priced laptops. The visualizations also showed differences in laptop availability across companies and laptop types. The correlation analysis helped identify relationships between numerical features, while the different distribution plots provided a clearer understanding of price variation and potential outliers.

Key Insights

1. The average laptop price in the dataset is approximately 59,870.
2. The median laptop price is approximately 52,055.
3. Laptop prices show considerable variation, with a standard deviation of approximately 37,243.
4. The dataset contains laptops ranging from approximately 9,271 to 324,955 in price.
5. The mean price being higher than the median indicates that higher-priced laptops influence the overall price distribution.
6. The company-distribution visualization shows that laptop records are not evenly distributed across all manufacturers.
7. The laptop-type visualization shows that some laptop categories are more common than others.
8. The price-distribution histogram shows that laptop prices are spread across a wide range rather than concentrated at a single price.
9. The correlation heatmap helps identify numerical features that have stronger relationships with laptop price.
10. The box plot and violin plot show substantial variation in laptop prices and help identify differences in price distributions and possible outliers.

Conclusion

This project analyzed a laptop dataset using Python and exploratory data analysis techniques. The dataset was loaded, explored, and cleaned before applying statistical analysis and multiple data visualizations. The analysis helped identify patterns in laptop prices, laptop types, and company distributions, while statistical measures provided a clearer understanding of price variation. The visualizations and findings provide useful insights into the characteristics of the laptop market and demonstrate how data analysis can be used to understand and interpret real-world datasets.

Challenges Faced

During the project, some challenges were encountered while working with the dataset and analysis process. These included understanding the dataset structure, handling unnecessary columns during data cleaning, preparing the data for analysis, and creating different visualizations. Managing the dataset correctly in Google Colab and organizing the code, outputs, and screenshots for documentation also required careful attention.

